

Wasiur Rahman Khuda Bukhsh

School of Mathematical Sciences
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Professional Experience

- Oct 2021 – **Assistant Professor**, *School of Mathematical Sciences*, The University of Nottingham, University Park, Nottingham, NG7 2RD, UK
- Jan 2020 – **President's Postdoctoral Scholar**, *Mathematical Biosciences Institute and the College of Public Health*, The Ohio State University, Columbus, Ohio, USA
- Oct 2018 – **Postdoctoral Researcher**, *Mathematical Biosciences Institute and the College of Public Health*, The Ohio State University, Columbus, Ohio, USA
- Jan 2014 – **Research Associate**, *Technische Universität Darmstadt*, Darmstadt, Germany
- Sep 2018 – Department of Electrical Engineering and Information Technology
- Jun 2011 – **Manager**, *ICICI Bank*, Mumbai, India
- Dec 2013 – Advanced Analytics, Business Intelligence Unit (BIU)

Education

- 2023 – **Fellow (FHEA)**, *Advance HE (formerly Higher Education Academy)*, UK
- 2023 – 2024 **Postgraduate Certificate in Higher Education (PGCHE)**, *University of Nottingham*, UK
- 2014 – 2018 **Ph.D. (Dr. rer. nat.)**, *summa cum laude*, *Department of Electrical Engineering and Information Technology*, *Technische Universität Darmstadt*, Germany
- 2009 – 2011 **Master of Statistics (M.Stat.)**, *Indian Statistical Institute*, Kolkata, India
- 2006 – 2009 **Bachelor of Science (B. Sc.) with Honours in Statistics**, *University of Calcutta*, Kolkata, India

Publications

Preprints/Submitted

- [1] Olga Izyumtseva and Wasiur R. KhudaBukhsh. Self-intersection local times for Volterra Gussian processes in stochastic flows with interaction. 2026. Submitted.
- [2] J. D. Harborne, Wasiur R. KhudaBukhsh, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Submitted.
- [3] Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and Wasiur R. KhudaBukhsh. Kernel-based estimators for functional causal effects. 2026. Under review.
- [4] Wasiur R. KhudaBukhsh and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.
- [5] Karim S. Elsayed, Olga Izyumtseva, Wasiur R. KhudaBukhsh, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.

- [6] Olga Izyumtseva and Wasiur R. KhudaBukhsh. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Revision requested.
- [7] Arnab Ganguly and Wasiur R. KhudaBukhsh. Enzyme kinetic reactions as interacting particle systems: Stochastic averaging and parameter inference. 2026. Revision requested.

Peer-reviewed journal publications

- [8] Arnab Ganguly and Wasiur R. KhudaBukhsh. Asymptotic analysis of the total quasi-steady state approximation for the michaelis–menten enzyme kinetic reactions. *Journal of Mathematical Analysis and Applications*, 561(1):130551, 2026.
- [9] Yi Fu, Hye-Won Kang, Wasiur R. KhudaBukhsh, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi-Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 1(2):202–228, 2026.
- [10] Riccardo Corradin, Luca Danese, Wasiur R. KhudaBukhsh, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.
- [11] Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, Wasiur R. KhudaBukhsh, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.
- [12] Yushuf Sharker, Zaynab Diallo, Wasiur R. KhudaBukhsh, and Eben Kenah. Pairwise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.
- [13] Wasiur R. KhudaBukhsh and Grzegorz A. Rempała. How to *correctly* fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.
- [14] Matthew Wascher, Patrick Schnell, Wasiur R. KhudaBukhsh, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.
- [15] István Z. Kiss, Luc Berthouze, and Wasiur R. KhudaBukhsh. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.
- [16] Wasiur R. KhudaBukhsh, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.
- [17] Wasiur R. KhudaBukhsh, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in ohio. *Journal of Theoretical Biology*, 561:111404, 2023.
- [18] Colin Klaus, Matthew Wascher, Wasiur R. KhudaBukhsh, and Grzegorz Rempała. Likelihood-Free Dynamical Survival Analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.
- [19] Kai Cui, Wasiur R. KhudaBukhsh, and Heinz Koepl. Hypergraphon mean-field games. *Chaos*, 2022.
- [20] Wasiur R. KhudaBukhsh, Casper Woroszylo, Grzegorz Rempała, and Heinz Koepl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.

- [21] Colin Klaus, Matthew Wascher, Wasiur R. KhudaBukhsh, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 5 2022.
- [22] Francesco Di Lauro*, Wasiur R. KhudaBukhsh*, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-markovian epidemic models. *Journal of the Royal Society Interface*, 2022. *Both authors contributed equally and are joint first authors.
- [23] Kai Cui, Wasiur R. KhudaBukhsh, and Heinz Koepl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 4 2022.
- [24] Harley Vossler, Pierre Akilimali, Yuhan Pan, Wasiur R. KhudaBukhsh, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: Study of 2018-2020 ebola outbreak in democratic republic of the congo. *Scientific Reports*, 12, 2022.
- [25] Ido Somekh*, Wasiur R. KhudaBukhsh*, Elisabeth Dowling Root*, Greg Rempala, Eric Simoes, and Eli Somekh. Quantifying the Population-level Effect of COVID-19 Mass Vaccination Campaign in Israel: A Modeling Study. *Open Forum Infectious Diseases*, 2022. *Equal contribution.
- [26] Wasiur R. KhudaBukhsh*, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. (*Invited paper).
- [27] Wasiur R. KhudaBukhsh, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.
- [28] Wasiur R. KhudaBukhsh, Arnab Auddy, Yann Disser, and Heinz Koepl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 9 2019.
- [29] Hye-Won Kang*, Wasiur R. KhudaBukhsh*, Heinz Koepl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. *joint first authors.
- [30] Saumya Yashmohini Sahai, Saket Gurukar, Wasiur R. KhudaBukhsh, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A Machine Learning Model for Nowcasting Epidemic Incidence. *Mathematical Biosciences*, 2021.
- [31] Wasiur R. KhudaBukhsh, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koepl. Generalized cost-based job scheduling in very large cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.
- [32] Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, Wasiur R. KhudaBukhsh, Joseph Tien, Marcel Yotebieng, Eben Kenah, and Grzegorz A. Rempała. Modeling Outbreak Data: Analysis of a 2012 Ebola Virus Disease Epidemic in DRC. *BIOMATH*, 8(2), 2019.
- [33] Wasiur R. KhudaBukhsh, Amr Rizk, Sounak Kar, and Heinz Koepl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 3 2019.
- [34] Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein, Robin Klose, Roland Kluge, Heinz Koepl, Boris Koldehofe, Wasiur R. KhudaBukhsh, Mahdi Mousavi, Martin Pfannmueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: A protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 4 2019.

Book chapters

- [35] Olga Izyumtseva, Wasiur R. KhudaBukhsh, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. *Handbook of Statistics*. Elsevier, 2024.
- [36] Grzegorz A. Rempała and Wasiur R. KhudaBukhsh. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

Peer-reviewed conference proceedings

- [37] Karim Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. On the Trade-off between Fidelity and Latency for the Quantum Link Layer with few Memories and Entanglement Purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)*, 2024. Best paper award.
- [38] Karim Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. On the Fidelity Distribution of Link-level Entanglements under Purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.
- [39] Riccardo Corradin, Luca Danese, Wasiur KhudaBukhsh, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.
- [40] Wasiur R. KhudaBukhsh, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koepl. Collaborative uploading in heterogeneous networks: Optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 4 2018. < 20% acceptance rate. Best-in-Session Presentation Award.
- [41] Wasiur R. KhudaBukhsh, Amr Rizk, Alexander Frömmgen, and Heinz Koepl. Optimizing Stochastic Scheduling in Fork-Join Queueing Models: Bounds and Applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 5 2017. ~ 20% per cent acceptance rate.
- [42] Adrian Šošić, Wasiur R. KhudaBukhsh, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, May 2017. Available: <http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf>.
- [43] Adrian Šošić, Wasiur R. KhudaBukhsh, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 5 2017. ~ 26% per cent acceptance rate, Best Paper Award Finalist.
- [44] Wasiur R. KhudaBukhsh, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koepl. Analysing and Leveraging Client Heterogeneity in Swarming-based Live Streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 5 2016. ~ 26% per cent acceptance rate.
- [45] Mahdi Mousavi, Hussein Al Shatri, Wasiur R. KhudaBukhsh, Heinz Koepl, and Anja Klein. Cross-Layer QoE-based Incentive Mechanism for Video Streaming in Multi-Hop Wireless Networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 9 2017.

Thesis and technical notes

- [46] Wasiur R. KhudaBukhsh. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: <http://tuprints.ulb.tu-darmstadt.de/7588/>.
- [47] Mark Sinzger-D'Angelo, Heinz Koepl, and Wasiur R. KhudaBukhsh. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: <https://arxiv.org/abs/1808.00258>.

Awards and Funding

Awards

1. Best Paper Award at the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)
2. 2019 President's Postdoctoral Scholars Program (PPSP) of the Ohio State University
3. Best-in-Session Presentation Award at the IEEE International Conference on Computer Communications (IEEE INFOCOM), Honolulu, Hawaii, USA, 2018.

Funding

1. Engineering and Physical Sciences Research Council (EPSRC): EP/Y027795/1 (79,987.58 GBP, 01/03/2024 - 28/02/2025)

Other smaller grants and travel awards

1. Isaac Newton Institute (INI) Retreats, June 29 - July 3, 2026. (Jointly with Olga Iziumtseva, and Ercan Sönmez)
2. Isaac Newton Institute (INI) Retreats, July 29 - August 4, 2024.
3. Center for Mathematical Machine Learning and its Applications (CM2LA) Seorabol Research Station Workshop grant, South Korea, June, 2025. (7,000 USD)
4. LMS Research in Pairs grant 2024 (1200 GBP)
5. MATRIX-Simons Travel Grant (2,500 AUD, 02/2024)
6. UoN Faculty of Science International Research Collaboration Fund (3,150 GBP)
7. Heilbronn Institute for Mathematical Research (HIMR) Small Grants Scheme (1,000 GBP)
8. Royal Society International Exchange grant 2022 (11,702.09 GBP)
9. UoN Faculty of Science International Collaboration Fund (4,375 GBP)
10. LMS Undergraduate Research Bursary 2022 (900 GBP + matching 900 GBP from the University of Nottingham)
11. LMS Research in pairs grant 2022 (1200 GBP)
12. Institute of Mathematical Statistics New Researchers Conference 2022 (1860 USD)
13. American Institute of Mathematics (AIM) SQuaREs (Structured Quartet Research Ensembles) (along with Hye-Won Kang, Lea Popovic, Greg Rempala, Ruth Williams and Felipe Campos)
14. The German Academic Exchange Service (DAAD) travel grant (Kongressreise) to participate in the Annual Meeting of the SMB in Sydney, Australia, 2018 (~ 2000 EUR)

Presentations

Summary: 58 invited and 30 contributed/local talks. 7 poster presentations.

Invited talks

1. Invited speaker, Workshop on "Individual variation and its consequences for population dynamics", University of Strathclyde, Glasgow, UK, May 6-7, 2026
2. Probability Seminar, Louisiana State University, 9 April, 2026
3. Mathematics in Life Sciences Seminar, University of Strathclyde, Scotland, UK, 18 March, 2026
4. Department Seminar, School of Mathematical and Statistical Sciences, IIT Mandi, 12 January 2026
5. International Conference on Recent Advances in Medical Science and Technology 2025, IIT Kharapgur, India, December 2025
6. HEALMOD Seminar (online), The Ohio State University, USA, November 25, 2025
7. Statistics and Probability seminar, University of Manchester, UK, October 22, 2025
8. Workshop on Gaussian and Levy-type stochastic processes, University of Nottingham, September 10, 2025.
9. Annual Meeting of the Royal Statistical Society, Edinburgh, UK, September 2-5, 2025
10. Stochastic Processes and their Applications 2025, Wrocław, Poland, 14-18 July 2025
11. MIDAS Webinar (online), June 27, 2025

12. BIRS-CMO Workshop on "Mathematical and Statistical Challenges in Post-Pandemic Epidemiology and Public Health", Oaxaca, Mexico, June 15-20, 2025
13. Statistics and Probability Seminar, University of Essex, UK, May 15, 2025
14. Innovations in Machine Learning, Artificial Intelligence, Data Science & Modelling (IMLAIDSM'24), India, December 2024
15. Probability Seminar, Louisiana State University, 25 November, 2024
16. Mathematics of Reaction Networks (MoRN) seminar, November 7, 2024
17. CMMB Seminar, University of Nottingham, October 22, 2024
18. Mini-symposium talk at the Annual Meeting of the Society for Mathematical Biology, Seoul, South Korea, June 30- July 5, 2024
19. Epidemiology workshop, Konkuk University, Seoul, South Korea, June 29, 2024.
20. Mathematical Biology Seminar, University of Leeds, December 7, 2023.
21. Mathematical Life Sciences talk, University of Bonn, Germany, December 1, 2023.
22. Probability Seminar, Louisiana State University, USA, November 7, 2023.
23. Mini-symposium speaker, ICIAM, Tokyo, Japan, August 20-25, 2023.
24. SMB Diversity in Math Bio seminar series, July 25, 2023.
25. Mini-symposium speaker, SIAM Conference on Optimization 2023, Seattle, Washington, USA, June 2, 2023.
26. BBSRC DTP event, University of Nottingham, UK, April 19, 2023.
27. Probability Seminar, Louisiana State University, USA, April 3, 2023.
28. Stochastics Seminar, Georgia Tech, USA, March 2, 2023. (online)
29. NANT Lecture, Networking in Applied Network Theory, Bedlewo, Poland, February 23, 2023. (online)
30. Theoretical Statistics and Mathematics Seminar, Indian Statistical Institute, New Delhi, December 14, 2022.
31. Invited speaker, Workshop on Non-equilibrium Phenomena in Physics and Biology, Asia Pacific Center for Theoretical Physics, South Korea, December 5-8, 2022.
32. Invited speaker, Self-Organizing Systems Workshop, Technische Universität Darmstadt, Germany, December 2, 2022.
33. Mini-symposium speaker at SIAM Conference on Uncertainty Quantification, April 15, 2022.
34. Statistics and Probability Seminar, University of Nottingham. February 3, 2022.
35. Mathematics of Reaction Networks (MoRN) seminar, January 27, 2022.
36. Math Bio seminar, The Ohio State University, virtual, November 4, 2021
37. Stochastic Systems seminar talk, Department of Mathematics, University of California San Diego, virtual, October 21, 2021
38. Guest lecture on infectious disease epidemiology, The Ohio State University, virtual, September 16, 2021
39. American Institute of Mathematics (AIM) Workshop on "Limits and control of stochastic reaction networks", July 28, 2021
40. Colloquium at the Department of Mathematics, Simon Fraser University, virtual, April 23, 2021
41. Special session at the Joint Mathematics Meetings (JMM), virtual, January 6-9, 2021
42. IDI Virtual COVID-19 Symposium, December 3, 2020
43. Seminar, Institute of Applied Mathematics and Mechanics, University of Warsaw, November 30, 2020
44. Seminar, Institute of Applied Mathematics and Mechanics, University of Warsaw, November 23, 2020
45. Biostatistics seminar, the Ohio State University, Columbus, OH, USA, October 30, 2020
46. Math-Bio seminar, Virginia Tech, virtual, September 23, 2020
47. International Webinar Series on Artificial Intelligence and Machine Learning, virtual, September 18, 2020
48. Workshop on "Life on Planet Earth: Above and Below", MBI, OSU, August 11, 2020
49. International Webinar, Departments of Mathematics and Physics, Bidhan Chandra College, virtual, July 18, 2020
50. Special session of Data Science, Business Intelligence Unit, ICICI Bank, Mumbai, India, July 24, 2020
51. Special session at the Erdős Institute Python Bootcamp, May 2020 (jointly with Marissa Renardy)

52. Biostatistics Seminar, the Ohio State University, Columbus, OH, USA, February 28, 2020
53. Mathematical Modeling and Statistical Analysis of Infectious Disease Outbreaks, CIRM, Marseille, France, February 21, 2020
54. Mini-symposium speaker at the Annual Meeting of the SMB, Montreal, Canada, 2019
55. Mini-symposium speaker at the International Conference on Mathematical Methods and Models in Biosciences (BioMath), June 2019, Poland
56. BIRS-CMO workshop on “Scaling Limits of Dynamical Processes on Random Graphs”, May 2019, Oaxaca, Mexico
57. Mini-symposium speaker at the Annual Meeting of the SMB and the JSMB, Sydney, Australia, 2018
58. Cloud Computing, Machine Learning And Networking Research (CLAN) Lab, Purdue University (May, 2017), Host : Vaneet Aggarwal

Contributed talks

1. MIDAS Webinar (online), December 11, 2026
2. Stochastic Processes and their Applications 2026, Ithaca, New York, USA, 15-19 June 2026
3. Annual Meeting of the Royal Statistical Society, Edinburgh, UK, September 2-5, 2025
4. Melbourne Mathematical Biology Seminar, University of Melbourne, Melbourne, Australia, June 20, 2024
5. IEEE International Conference on Communications (IEEE ICC), Denver, Colorado, USA, June 9 - June 14, 2024
6. Mini-symposium talk at the Annual Meeting of the Society for Mathematical Biology, Columbus, OH, USA, July 16-21, 2023
7. Dynamical Systems in Life Sciences, The Ohio State University, July 13-15, 2023
8. BIRS Workshop on “Preparing for the next pandemic”, June 12-17, 2022
9. Annual Meeting of the Society for Mathematical Biology, June 13-17, 2021
10. Workshop on Mathematical and Computational Biology, June 10-11, 2021
11. The SIAM Conference on Applications of Dynamical Systems (DS21), May 23-27, 2021
12. ENAR Spring Meeting, March 14-17, 2021
13. Mathematical Biosciences Institute (MBI) Seminar, Columbus, Ohio, USA, March 04, 2021
14. Joint Mathematical Epidemiology and Math Education SMB Subgroup Meeting, Feb 7-8, 2021
15. Applied Mathematics Seminar, Department of Mathematics, The Ohio State University, November 19, 2020
16. Bernoulli–IMS One World Symposium 2020, virtual, August 25, 2020
17. Mini-symposium speaker at the Annual Meeting of the SMB, virtual, August 19, 2020
18. Joint Statistical Meetings 2020, virtual, August 2-6, 2020
19. Mini-symposium speaker at SIAM Conference on Mathematics of Data Science 2020 (virtual), June 2020
20. MBI Workshop on Mathematical and Computational Methods in Biology, May 6, 2020
21. International Indian Statistical Association Conference (IISA 2019), Mumbai, India, December 28, 2019
22. Latin American Congress of Probability and Mathematical Statistics (CLAPEM) XV, Merida-Yucatán, México, December, 2019
23. Mathematical Biosciences Institute (MBI) Seminar, Columbus, Ohio, USA, October, 2019
24. Joint Statistical Meetings (JSM) - American Statistical Association, Colorado, USA, 2019
25. Stochastic Processes and their Applications, Bernoulli Society, Northwestern University, July 2019, USA
26. International Conference on Mathematical Methods and Models in Biosciences (BioMath), June 2019, Poland
27. Mathematical Biosciences Institute (MBI) Seminar, Columbus, Ohio, USA, 2019
28. IEEE International Conference on Computer Communications (IEEE INFOCOM), Honolulu, Hawaii, USA, 2018
29. IEEE International Conference on Computer Communications (IEEE INFOCOM), Atlanta, GA, USA, 2017
30. IFIP Networking Conference and Workshops, Vienna, Austria, 2016

Poster presentations

1. Annual Meeting of the Society for Mathematical Biology, Heidelberg, September 19-23, 2022
2. 22nd Meeting of New Researchers in Statistics and Probability, George Mason University, USA, August 4, 2022
3. Mathematical Models in Evolutionary Biology conference, CIRM, Marseilles, France, February 10-14 2020
4. Annual Meeting of the Infectious Disease Institute, The Ohio State University, September 2019, Columbus, OH, USA
5. Joint Statistical Meetings (JSM) - American Statistical Association, Colorado, USA, 2019
6. CMCF Algorithms and Methods for Single Cell Genomics, University of California, Irvine, June 2019, USA
7. BIRS-CMO workshop on "Scaling Limits of Dynamical Processes on Random Graphs", May 2019, Oaxaca, Mexico

Teaching Experience

University of Nottingham, UK

1. MATH3060 Financial Mathematics stream,
School of Mathematical Sciences, The University of Nottingham
Role: Instructor
Terms: Academic year 2025-26 (jointly with Michael Tretyakov)
2. MATH2109: Probability 3/Markov Chains
School of Mathematical Sciences, The University of Nottingham
Role: Instructor
Terms: Spring 2025 (~250 students), Spring 2026 (~270 students)
3. MATH2010: Probability Methods and Models
School of Mathematical Sciences, The University of Nottingham
Role: Instructor
Terms: Spring 2022 (~270 students), Spring 2023 (~260 students), Spring 2024 (~282 students)
4. MATH4045 Mathematical Finance stream,
School of Mathematical Sciences, The University of Nottingham
Role: Instructor
Terms: Academic year 2022-23 (jointly with Fabrizio Leisen), Academic year 2023-24 (jointly with Fabrizio Leisen in autumn & John Billingham in spring), Academic year 2024-25 (jointly with Marco Iglesias)

The Ohio State University, USA

1. STAT 4201: Introduction to Mathematical Statistics
Department of Statistics, The Ohio State University, USA
Role: Instructor
Terms: Fall semester 2019 (> 120 students)
2. PUBHBIO7193: Individual studies in Biostatistics (Advanced stochastic epidemiology)
Division of Biostatistics, The Ohio State University, USA
Role: Instructor (jointly with Eben Kenah)
Terms: Summer semester 2019

Technische Universität Darmstadt, Germany

1. Computational Methods for Systems and Synthetic Biology (CMSSB)
Department of Electrical Engineering and Information Technology, Technische Universität Darmstadt, Germany
Role: Teaching Assistant
Terms: Summer semester 2016, Summer semester 2017

Supervision and Mentoring

Summary: Supervised/Co-supervised a total of 16 undergrad (B.Sc., not including REU students) and 17 M.Sc. students.

Postdoctoral Researchers/Fellowships

1. Dr. Olga Iziuntseva (British Academy Fellowship), 09/2023 - 08/2026, School of Mathematical Sciences, University of Nottingham, UK
2. Dr. Yangrui Xiang, 09/2024 - 06/2025, School of Mathematical Sciences, University of Nottingham, UK

PhD Students

1. Kushankur Dutta, 10/2023 - 04/2027 (expected), School of Mathematical Sciences, University of Nottingham, UK (Secondary supervisor: Mikhail Tretyakov and Karthik Bharath)
2. Wesam El Buaishi, 06/2023 - 12/2026 (expected), School of Mathematical Sciences, University of Nottingham, UK (Primary supervisor: Yordan Raykov)
3. James Harborne, 10/2022 - 04/2026 (expected), School of Mathematical Sciences, University of Nottingham, UK (Primary supervisor: John King)

M.Sc. Students

1. Jennifer Wood, 2026, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
2. Wenhao Qiu, 2025, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
3. Maxwell Hancock, 2025, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
4. William Bateman, 2025, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
5. Siyu Xu, 2024, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
6. Qi Zhang, 2024, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
7. Kaustav Chakraborty, 2024, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
8. Nazreen Asok, 2023, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
9. Changqing Du, 2023, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
10. Yajie Guo, 2022, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
11. Jianing Ye, 2022, Master degree dissertation, School of Mathematical Sciences, University of Nottingham, UK
12. Robert Lumpkin, 2019, Master degree project (jointly with Prof. Greg Rempala), Department of Mathematics, The Ohio State University, USA (*resulted in the publication [32]*)
13. Arnab Auddy, 2017, Summer internship, Technische Universität Darmstadt, Germany (*resulted in the publication [28]*)
14. Markus Schanz, 2017, Master thesis (jointly with Christian Koch), Technische Universität Darmstadt, Germany
15. Hameer Abbasi, 2017, Master thesis (jointly with Bastian Alt), Technische Universität Darmstadt, Germany
16. Ranjani Krishnan, 2018, Master thesis (jointly with Denny Stohr), Technische Universität Darmstadt, Germany
17. Sayantan Kumar, 2018, Summer internship, Technische Universität Darmstadt, Germany

B.Sc. Students

1. Yiwen Yan, 2025, Summer internship, School of Mathematical Sciences, The University of Nottingham, UK
2. Maoduan Ran, 2023, Summer internship, School of Mathematical Sciences, The University of Nottingham, UK
3. Sarina Rivlin-Sanders, 2022, Summer internship, School of Mathematical Sciences, The University of Nottingham, UK
4. Junzhe Zhu, 2022, Summer internship, School of Mathematical Sciences, The University of Nottingham, UK
5. Jasen Lai, 2019, Summer project, The Ohio State University, USA
6. Ran An, 2019, Summer project, The Ohio State University, USA
7. Stanley L. Cao, 2019, Summer project, The Ohio State University, USA
8. Hongyi Wang, 2019, Summer project, The Ohio State University, USA
9. Joey Zeng, 2019, Summer project, The Ohio State University, USA
10. Kai Li, 2019, Summer project, The Ohio State University, USA
11. Yi Dai, 2019, Summer project (jointly with Prof. Greg Rempala and Prof. Eben Kenah), The Ohio State University, USA
12. Jordan Kirsch, 2019, Summer project (jointly with Prof. Greg Rempala and Prof. Eben Kenah), The Ohio State University, USA
13. Vikash Vikash, 2016, Summer internship, Technische Universität Darmstadt, Germany
14. Simon Schwanz, 2016, Proseminar, Technische Universität Darmstadt, Germany
15. Simon Schwanz, 2017, Project seminar (jointly with Christian Koch), Technische Universität Darmstadt, Germany
16. Siddhaarth Sarkar, 2018, Summer internship, Technische Universität Darmstadt, Germany

Outreach and Service

Internal Examiner of PhD Theses

1. Liam Critcher, School of Mathematical Sciences, University of Nottingham, UK, 2025.

External Examiner of PhD Theses

1. Hiu Ching Yip, Politecnico di Torino, Italy, 2024.
2. Dániel Keliger, Department of Stochastics, Institute of Mathematics, Budapest University of Technology and Economics, Hungary, 2025.

Editorial Service

1. Guest Associate Editor (On behalf of the Applied Probability Section of the RSS), Discussion paper for the Journal of the Royal Statistical Society Series B, 2023.
2. Guest editor, special issue of Mathematical Biosciences and Engineering (MBE).

Grant Application Review

1. Engineering and Physical Sciences Research Council (EPSRC), UK Research and Innovation (UKRI), 2024.
2. The Netherlands Organisation of Health, Research and Development (ZonMw), 2024.

Professional committees

1. Committee of the Applied Probability Section of the Royal Statistical Society (RSS). 2023 - .
2. Secretary and Treasurer, Reaction Networks Subgroup of the Society for Mathematical Biology (SMB), 2025 - .

COVID-19 Modelling Response

1. Provided service to the Ohio Department of Health and Ohio Hospital Association as part of the OSU/IDI COVID-19 response modelling team. Provided daily predictions of COVID-19 cases and resultant hospital

burden in the state of Ohio, USA.

Service to the Department/University

1. Course Director, B.Sc./M.Math. with a year in industry, University of Nottingham, 2022- .
2. Organizer of the Statistics and Probability Seminars, 2022 - 2023.
3. Reviewer for the President's Research Excellence, The Ohio State University, 2021.
4. Judge for the OSU Mathematical Contest for Modeling, The Ohio State University, 2018, 2019.
5. Judge for OPA travel awards, the Ohio State University, 2019.

Organizing conferences

1. Stochastic Reaction Networks subgroup Mini-symposium at the Annual Meeting of European Society for Mathematical and Theoretical Biology (ESMTB) the Society for Mathematical Biology (SMB), July 13-18, 2026 at Graz, Austria (jointly with Stefan Müller).
2. Mini-symposium at the Annual Meeting of European Society for Mathematical and Theoretical Biology (ESMTB) the Society for Mathematical Biology (SMB), July 13-18, 2026 at Graz, Austria (jointly with Olga Iziomtseva and Arnab Ganguly).
3. Contributed session at the Stochastic Processes and their Applications Conference 2026, Cornell University, June 14-20, 2026, USA. (jointly with Olga Iziomtseva)
4. Workshop on "Recent advances in stochastic processes: theory and applications", University of Edinburgh, UK, May 27, 2026 (jointly with Tess Vernon)
5. Workshop on "Gaussian and Levy-type stochastic processes", University of Nottingham, UK, September 10, 2025 (jointly with Olga Iziomtseva)
6. BIRS Workshop on "Mathematical and statistical challenges in post-pandemic epidemiology and public health", June 15-20, 2025 (jointly with Jessica Stockdale, Sara Del Valle, Joel C. Miller and Grzegorz Rempala)
7. Satellite Workshop on Chemical Reaction Networks, Gyeong Ju, South Korea, July 6-9, 2024 (jointly with Hye-Won Kang, and Jinsu Kim)
8. Mini-symposium at the Annual Meeting of the Society for Mathematical Biology, Seoul, South Korea, June 30- July 5, 2024 (jointly with Hye-Won Kang and Jinsu Kim)
9. Workshop on Epidemic Modelling on behalf of the Applied Probability Section of the Royal Statistical Society (RSS), October 11, 2023 (jointly with Alex Cox, Fraser Daly, and Lucy Teese)
10. Mini-symposium at the Annual Meeting of the SMB, Columbus, Ohio, July 2023. (Jointly with Hye-Won Kang.)
11. Mini-symposium at SIAM Optimization, Seattle, Washington, USA, May 2023.
12. Workshop on "Recent challenges in mathematical epidemiology" at University of Nottingham, UK, September 27, 2022. (jointly with Kirsty Bolton and Yordan Raykov)
13. Invited session at the IMS Asia Pacific Rim Meeting, Melbourne, Australia, 2022 (pending acceptance. Jointly with Gursharn Kaur)
14. Mini-symposium at the ECMTB 2022, September 19-23, 2022 (jointly with Hye-Won Kang)
15. BIRS Workshop on "Preparing for the next pandemic", June 12-17, 2022 (jointly with Sara Del Valle, Joel C. Miller and Rick Durrett)
16. Mini-symposium at the SIAM Conference on Applications of Dynamical Systems, 2021
17. Mini-symposium at the Annual Meeting of the Society for Mathematical Biology, 2021 (Jointly with Hye-Won Kang)
18. Invited session at the Annual Meeting of the ENAR, 2021 (jointly with Marissa Renardy)
19. Mini-symposium at the Annual Meeting of the Society for Mathematical Biology, 2020 (jointly with Hye-Won Kang)
20. Mini-symposium at the SIAM Conference on Mathematics of Data Science (MDS20), Cincinnati, OH, USA, June 11, 2020 (jointly with Veronica Ciaconel)
21. Contributed Session at the Latin American Congress of Probability and Mathematical Statistics (CLAPEM) XV, Merida-Yucatán, México, December, 2019 (jointly with Hye-Won Kang)

22. Session at the 2019 International Indian Statistical Association Conference (IISA 2019), Mumbai, India, December 2019 (jointly with Arindam Fadikar)

Tutorials

1. "How to write scientific papers" for MBI REU students, 2019
2. "Introduction to epidemic modelling" for SAMMS REU students, 2019
3. "Modeling the COVID-19 pandemic" for the Erdős Institute bootcamp, May 2020

Mentoring REUs

1. MBI REUs, Summer 2019
2. Ohio 5 SURE students at the College of Public Health, OSU, 2019

Referee Service

1. Electronic Journal of Probability
2. Journal of Applied Probability/Advances in Applied Probability (Applied Probability Trust)
3. Stochastic Processes and their Applications
4. Stochastics (Taylor & Francis)
5. Statistics and Probability Letters
6. SIAM Journal on Applied Dynamical Systems
7. SIAM Multiscale Modeling & Simulation
8. Biometrika
9. Mathematical Biosciences
10. Bulletin of Mathematical Biology
11. Journal of Chemical Physics
12. Journal of Mathematical Biology
13. Journal of Theoretical Biology
14. PLOS One
15. PLOS Computational Biology
16. Theoretical Population Biology
17. Journal of Mathematical Physics
18. Mathematical Biosciences and Engineering
19. European Journal of Control
20. Afrika Matematika
21. Mathematics (MDPI)
22. IEEE Transactions on Automatic Control
23. Journal of Mathematics in Industry
24. Journal of Mathematical Analysis and Applications
25. Journal of Scientific Research
26. Methodology and Computing in Applied Probability
27. IEEE Control System Letters
28. IEEE Transactions on Networking
29. IEEE Transactions on Control of Network Systems
30. IEEE Multi-Conference on Systems and Control, Sydney, Australia, 2015
31. American Control Conference (ACC)

Review Services

1. AMS Mathematical Reviews (MathSciNet)
2. zbMATH Open

Other Services

1. Judge for abstract submissions at the Annual Meeting of the MIDAS Network, 2021
2. Judge for poster sessions at the Annual Meeting of the SMB 2021, 2023.

Professional Memberships

1. Royal Statistical Society (RSS) (Applied Probability Section Committee member)
2. The Society for Mathematical Biology (SMB) (Life member)

Computer Skills

Programming Languages C, R, Python, SQL, Julia

Software Matlab, SAS, Microsoft Office, $\LaTeX$

Languages

English Professional Proficiency

Bengali Mother Tongue

Hindi Working Knowledge

German Intermediate (B1 level)

Media highlights

1. The Dynamical Systems Web (DSWeb): Student Feature in the Jan 2021 edition of the journal. (URL: <https://tinyurl.com/dsweb>)
2. OSC Research Report: COVID-19 projection models equip Ohioans. (URL: <https://tinyurl.com/resrchrprt>)
3. OSC News: Pandemic Modeling: COVID-19 Projection Models Equip Ohioans (URL: <https://tinyurl.com/oscwasiur>)
4. HPC Wire: Pandemic Modeling: COVID-19 Projection Models Equip Ohioans (URL: <https://tinyurl.com/hpcnews>)
5. IDI News: Far from trivial: A postdoctoral researcher's path to a key position in the OSU/IDI COVID-19 modeling team (URL: <https://tinyurl.com/IDInews>)
6. STEAM Member Spotlight - Dr. Wasiur Rahman KhudaBukhsh (URL: https://www.youtube.com/watch?v=j_DEgauzw7w)

References

Available upon request.